

COD891 · END-TERM

Causal Knowledge Graph for Indian Ecosystem

Mechanism-first extraction from literature — towards a queryable substrate

Agriculture × Water Scarcity

End-Term Presentation · COD891

Knowledge graph · cross-stressor pathways · future Q&A platform

A causal query–answer platform over literature-backed evidence

District-style questions → knowledge graph traversals → answers with mechanisms + paper provenance

Q1

Which stressors are active for a district given some values of variables(ex rainfall, groundwater level, etc)?

IDEA

Traverse INDICATES-style links from observed variables → stressor families, using thresholds and indicator direction.

Q2

What is the full causal chain from root to symptom?

IDEA

Traverse CAUSES edges (consolidated_causal_chains); return mechanism text and provenance along the path.

Not prose summaries — a structured substrate where claims are traceable and reasoning is path-based: prerequisite for auditable decision support.

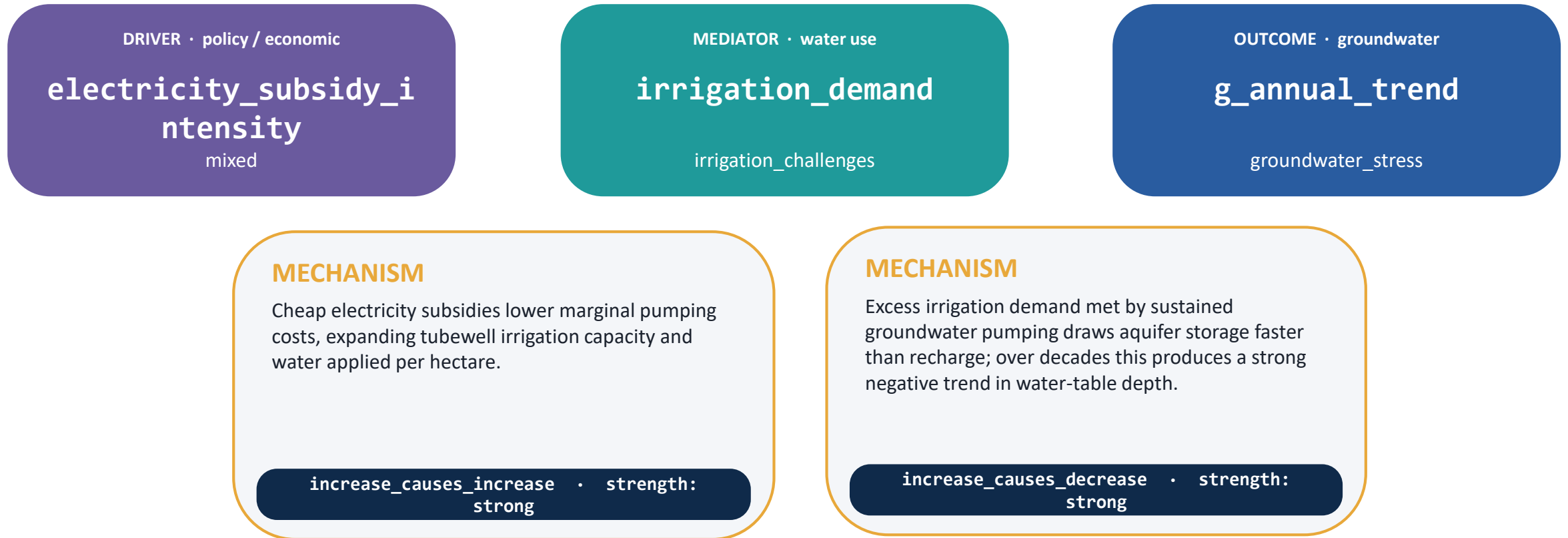
Community framing of stress: from production system to causal strands



What is in this submission — and what is the wider programme

Dimension	Programme (full)	This submission
Production system	Agriculture, Livestock, NTFP, Fishery, Human health, ...	Agriculture only
Observational stress	Multiple per system	Water scarcity
Causal decomposition	Many L2 strands per stress	Drought · Groundwater · Irrigation · Rainfed
Corpus	Intended ~24–30+ papers	14 papers (current master)

Two CAUSES edges, both cross-stressor — verbatim from the master



Subsidy lowers pumping cost → more irrigation water demanded → groundwater trend declines. Two CAUSES edges, multi-paper provenance.

What is a “variable”? — the canonical, mergeable unit

Defined by VARIABLE_TAXONOMY

variable_name

canonical snake_case id used on every edge
(from_var, to_var)

definition

what is measured · units · typical Indian data sources
(IMD, CGWB, MoA, RS)

qualifier rules

season, severity, medium are carried on the instance
(not baked into duplicate node names)

NEW_* concepts

papers may introduce unknowns; consolidation
proposes taxonomy_expansion_candidates

EXAMPLE · variable instance

salinity_level

role

context

cluster

rainfed_risk

medium

soil

season

kharif

severity

moderate

units

dS/m

data_source

ICAR/CSSRI · RS

alias

soil EC

One variable + one edge — the atoms of the graph

VARIABLE (consolidated_variables)

```
{
  "variable_name": "irrigation_demand",
  "role": "driver",
  "cluster_family": "irrigation_challenges",
  "definition": "agricultural water demand...",
  "units": "mm/season | m^3/ha",
  "qualifiers": { "season": "kharif" },
  "threshold_or_range": ">= 450 mm",
  "indicator_direction": "↑ → stress",
  "data_sources": ["MoA", "CGWB", "RS"],
  "supporting_papers": ["p03", "p07",
                        "p09", "p12"]
}
```

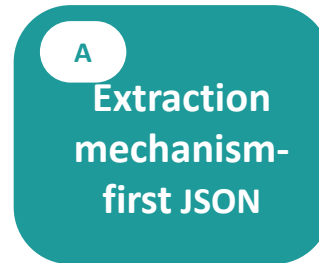
EDGE (consolidated_causal_chains)

```
{
  "edge_id": "e_master_42",
  "from_var": "irrigation_demand",
  "to_var": "g_annual_trend",
  "direction": "+",
  "strength": "high",
  "from_cluster_family": "mixed",
  "to_cluster_family": "groundwater_stress",
  "is_cross_stressor": true,
  "mechanism": "Higher seasonal demand ...
               draws aquifer beyond recharge",
  "mechanism_domains": ["hydrogeology", "agronomy"],
  "supporting_papers": ["p03", "p09"]
}
```

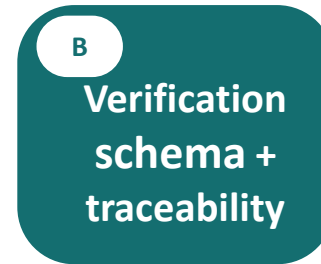
End-to-end pipeline



Selected from
corpus



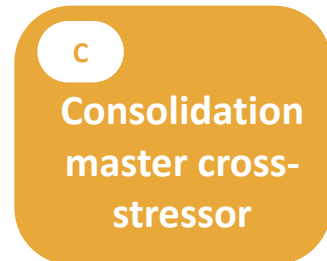
EXTRACTION



VERIFICATION

Iteration loop:

Re-extract → regenerate master → promote recurring NEW_* into VARIABLE_TAXONOMY → re-extract once → stop when duplicate-node and alias rates fall below thresholds.



CONSOLIDATION



JSON



search · paths
cross-stressor filter

Merge across papers · master_cross_stressor mode

What gets merged — and what is kept verbatim

Variables

single node per canonical name; merged thresholds; per-paper data_sources retained.

Edges

merge by (from_var, to_var); union mechanism_domains; supporting_papers list grows.

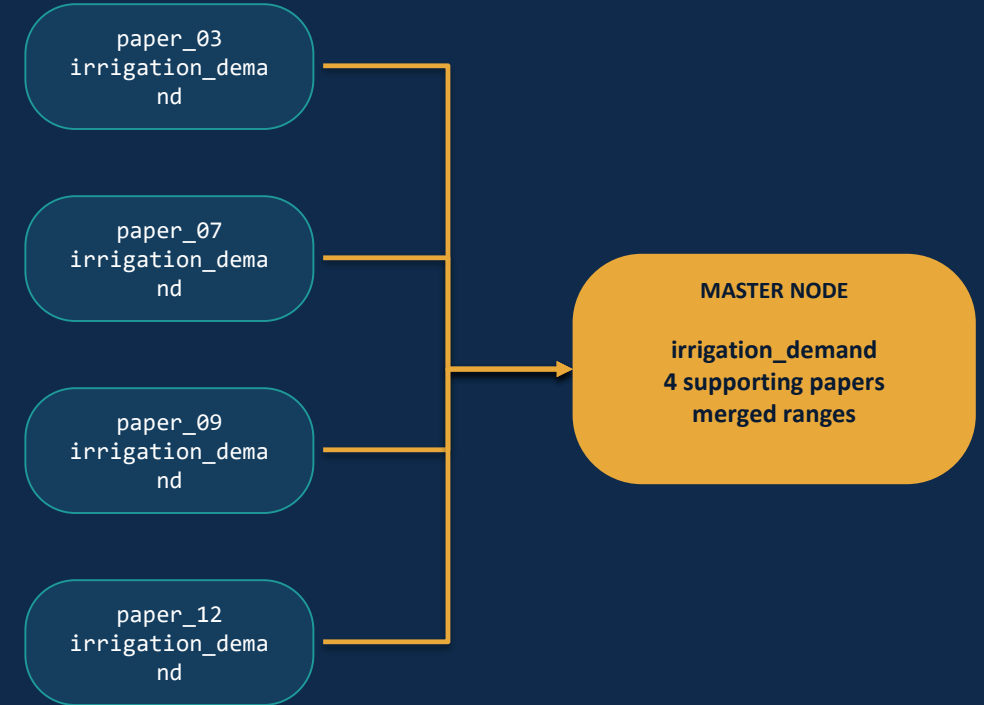
Cross-stressor

edges that bridge families are NOT stripped — kept and counted.

Aggregates

evidence_gaps, taxonomy_expansion_candidates, consolidation_summary, top cross-stressor cascades.

Many-paper extractions → one master node



The four arrays a query layer reads

consolidated_variables

GRAPH NODES

One object per canonical variable after cross-paper merge.

```
fields: merged_description · unit · role (driver / mediator / outcome /
context) · cluster_family · threshold_or_range · indicator_direction ·
data_sources · per_paper_ranges · paper_count · confidence
```

Defines nodes; future INDICATES / activation reads thresholds here.

consolidated_causal_chains

GRAPH EDGES

One object per directed causal edge in the merged graph.

```
fields: from_var · to_var · mechanism · direction · strength · conditions ·
from_/to_cluster_family · is_cross_stressor · mechanism_domains ·
supporting_papers · *_merged value ranges · *_by_paper for audit
```

What you traverse for full-causal-chain questions.

unified_mechanism_blocks

STORY VIEW

Named narrative bundles — primary_edges grouped.

```
fields: master_cross_stressor_cascade · primary_edges (logical refs to the
edge list) · optional secondary / context edges
```

Slide-friendly cascade view; coherent multi-step path.

new_variable_name_merges

AUDIT LOG

Records when consolidation collapses aliases into one canonical name.

```
fields: old_names → chosen_canonical · rationale · alias_resolution
decisions
```

Transparency: shows distinct concepts were not silently equated.

Master consolidation — 14 papers

14

papers

86

unique variables

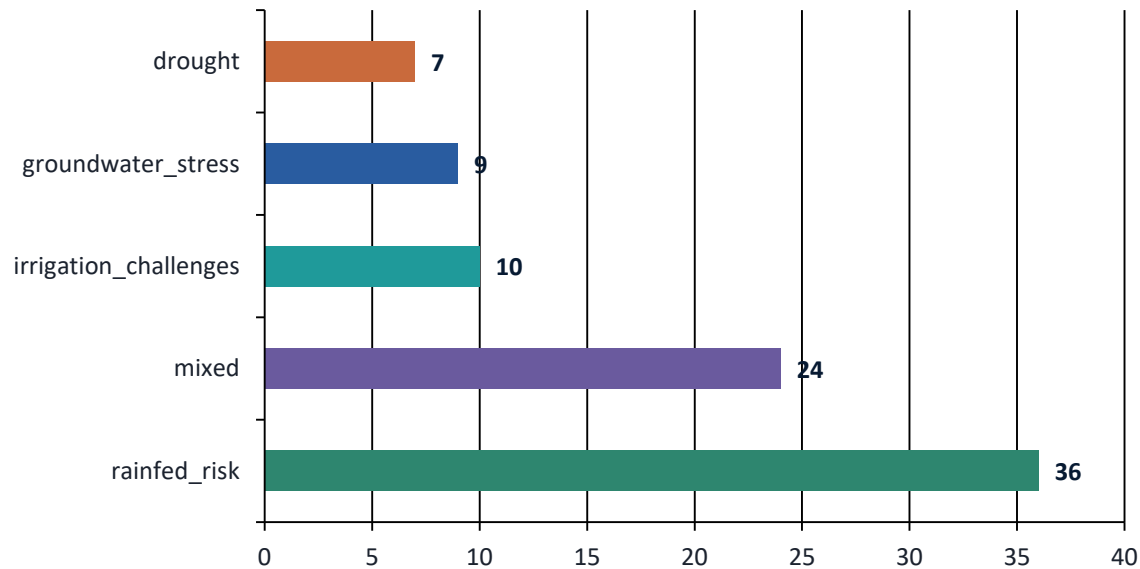
141

causal edges

60

cross-stressor edges (42.6%)

Variables by cluster_family



Variables by role



Headline: ~42.6% of edges bridge stressor families — the cross-stressor story is real and quantifiable.

Mechanism-first JSON (EXTRACTION_PROMPT)

- **Mechanism blocks first.** Model fills mechanism_blocks (named mechanisms with edge_drafts) before deduplicating.
- **Stable edge ids.** Edges acquire e1...en in causal_chains; mechanism_blocks reference these ids — no orphan duplicates.
- **Mandatory mapping.** Paper phrasing → canonical taxonomy name + alias_used_in_paper.
- **Cross-stressor flagged.** Edges carry from_cluster_family / to_cluster_family; is_cross_stressor when families differ.
- **Numeric grounding.** Where the paper reports values, threshold_or_range and indicator_direction travel with the variable.

Inside one paper

mechanism_blocks

pumping_dynamics
• edge_ids:
[e2,e5]

rainfall_runoff
• edge_ids:
[e2,e5]

policy_signal
• edge_ids:
[e2,e5]

causal_chains · stable edge ids

e1: rain_var →
soil_moist

e2: subsidy →
pumping

e5: pumping →
table_decline

Schema checklist

What the verification pass enforces

- ✓ Unique edge_id across the paper (e1...en)
- ✓ Every edge in causal_chains appears in some mechanism_blocks.edge_ids — and vice versa
- ✓ Provenance on every claim · paper_id · page / section
- ✓ Numeric ranges carry source citations
- ✓ Hybrid tags on nodes & edges: cluster_family + mechanism_domains
- ✓ Qualifier consistency (season, severity, medium) checked against rules
- ✓ schema_version bumped to 2.0 on the verified output

BEFORE · raw extraction

- edges live only inside mechanism_blocks
- duplicate edges between blocks
- mismatched alias names
- no schema_version stamp
- missing page numbers on quotes

AFTER · *_verified.json (schema 2.0)

- unique e_ids globally
- blocks reference edge_ids only
- aliases resolved to canonical
- schema_version: "2.0"
- every numeric range carries citation

How the system learns vocabulary — and what it links

expansion_candidates_canonical_seed.csv (one example row)

canonical_variable_name	NEW_electricity_subsidy_intensity
proposed_taxonomy_name	electricity_subsidy_intensity
cluster_families_reported	mixed · irrigation_challenges
roles_reported	driver
canonical_definition_master	share of agri power consumption at subsidised tariff
merge_notes	alias: 'free electricity coverage'
priority	HIGH

Top cross-stressor cascades (verbatim from master)

1	irrigation_demand → g_annual_trend mixed → groundwater_stress · high
2	NEW_electricity_subsidy_intensity → irrigation_demand mixed → irrigation_challenges · high
3	prec_annual_avg_rain → NEW_soil_moisture_index drought → rainfed_risk · high
4	NEW_land_holding_size → NEW_micro_irrigation_adoption mixed → irrigation_challenges · medium
5	NEW_rainwater_harvesting_coverage → recharge_rate mixed → groundwater_stress · medium

Limitations we will say first — and where we go next

LIMITATIONS

- ▶ **Manual pipeline.**
No single-command batch yet · no JSON-schema gate in Cl.
- ▶ **Less papers papers.**
Informative; evidence_gaps lists where literature is thin.
- ▶ **Q&A and INDICATES.**
Specified architecturally; not yet shipped.
- ▶ **Geographic binding.**
Mixed spatial focus across papers; block-level Q&A needs region linkage.
- ▶ **Too many variables**
Right now there are many variables, we need to prune it to keep the important once.

ROADMAP

Corpus

→ 24–30+ papers; target evidence_gaps.

Automation

Scripted extract → verify → consolidate · manifests · JSON Schema validation.

Query layer

Formal INDICATES + activation_score · path ranking · optional Neo4j/DuckDB load.

NL interface

LLM maps question → traversal + templated answer with citations.

Horizontal extension

Repeat for Livestock, Fishery, NTFP, Human health ...

THANK YOU

Questions · Discussion